Lukas Klein

(Dr. sc. ETH)

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Research Profile

My research focuses on AI systems that discover biomarkers capturing mechanistic and multimodal biological traits. It extends also into spatial omics foundation models, explainable AI, and therapy selection. I pursued this direction early on during my PhD at ETH Zürich and the German Cancer Research Center (DKFZ), supported by the competitive DKFZ fellowship, and now lead it jointly at EPFL and University Hospital Zürich, translating between computer science and biomedicine. This work has led to first-author publications by me and the students I supervise, in venues such as Nature Communications, Advanced Materials, or NeurIPS.

Professional Experience

École Polytechnique Fédérale de Lausanne (EPFL) *06/2025 - Today*
Postgraduate Researcher – In the AI for Molecular Medicine Group with Prof. Bunne.
- Collaborator in the National AI Initiative for Precision Oncology Innosuisse Flagship Grant.

University Hospital Zürich (USZ) *06/2025 - Today*
Postgraduate Researcher – In the Department of Oncology with Prof. Wicki.

Eidgenössische Technische Hochschule (ETH) Zürich *08/2021 - 06/2025*
Graduate Researcher – Supervised by Prof. Buhmann and Prof. Krause.

German Cancer Research Center (DKFZ) *08/2021 - 06/2025*
Graduate Researcher – Supervised by Dr. Jäger (DKFZ, now Latent Labs) and Prof. Maier-Hein (DKFZ).

Eidgenössische Technische Hochschule (ETH) Zürich *02/2021 - 08/2021*
Research Intern – Institute for Machine Learning supervised by Prof. Buhmann.

Camelot ITLab *01/2019 - 08/2019*
Consultant – Machine Learning and cloud solutions in the pharma and chemical industry.

Microsoft *06/2017 - 09/2017*
Intern – National Technology Office supervised by Thomas Langkabel.

Education

Eidgenössische Technische Hochschule (ETH) Zürich *2021 - 2025*
Ph.D. in Computer Science
- External group member in the IVIA Lab of Prof. El-Assady.
- Collaboration with the MIT-IBM Watson AI Lab supervised by Dr. Strobelt.

Helmholtz International Graduate School for Cancer Research *2021 - 2025*
DKFZ International PhD Program for Cancer Research

University of Oxford *2022*
OxML Summer School for Machine Learning in Healthcare.

University of Geneva *2019 - 2021*
M.Sc. in Statistics – Final grade: Very Good. Master Thesis and Exchange Semesters at ETH Zürich.

University of Mannheim *2015 – 2018*
B.Sc. in Economics – Final grade: Good. Exchange Semesters at the Norwegian School of Economics.

Academic Services

Invited Talks and Seminars

09/2026 [Talk] **AI+X Summit**
08/2026 [Seminar] **University of Basel** Pharmaceutical Sciences Summer School
06/2026 [Talk] **Wellcome Sanger** Artificial Intelligence for Cell Engineering (AICE) Lab

02/2026 [Talk] **German Diabetes Center (DDZ)** "Seeing Diabetes Differently" Symposium
10/2025 [Talk] **Forbeck Forum** "AI for Drug Discovery in Oncology"
11/2024 [Talk] **Tufts University** Visual Analytics Lab (VALT)
10/2024 [Talk] **MIT-IBM Watson AI Lab**
10/2024 [Talk] **Harvard University** Harvard Data Science Initiative (HDSI)
10/2024 [Talk] **Harvard University** Visual Computing Group
10/2024 [Talk] **MIT** Visualization Group
06/2024 [Talk] **IBM Research** AI for Scientific Discovery Group
12/2023 [Talk] **NeurIPS** XAI in Action 2023 Workshop
06/2023 [Talk] **TU Munich** Computational Imaging and AI in Medicine Group
06/2023 [Talk] **Helmholtz AI Conference** Interdisciplinary discussion "When do we blindly trust in AI?" (YouTube Video)
06/2023 [Seminar] **Helmholtz Imaging & Helmholtz AI Conference** Explainable AI for Discovery
04/2023 [Talk] **DESY Research Center**
09/2022 [Seminar] **Helmholtz Information & Data Science Academy (HIDA)** HIDA Summer School
05/2022 [Talk] **IJCAI** Workshop on Explainable AI 2022
05/2022 [Seminar] **HIDSS4Health Graduate School** Seminar for Analysing Medical Images Using ML

Reviewer

Nature (Main Journal)
Neural Information Processing Systems (**NeurIPS**, Top Reviewer award 2026)
International Conference on Machine Learning (**ICML**)
International Conference on Learning Representations (**ICLR**)
Eurographics Conference on Visualization (**EuroVis**)
Including various Conference Workshops.

Teaching

2025-2026 [Teaching Assistant] Foundation Models and Generative AI (EPFL)
2026 [Lecturer] Precision Oncology (University of Zürich)
2025 [Teaching Assistant] Machine Learning for Science (EPFL)
2023-2024 [Teaching Assistant] Advanced Topics in Machine Learning (ETH Zürich)
2022-2024 [Lecturer] Biomedical Imaging and Radiation Oncology (University of Heidelberg)

Mentorship and Supervision

2026 Allen Zhao – MSc Intern
Jointly supervised with Michael Moor (ETH Zürich).

2026 Anja Sjöström – Semester Project Student and MSc Research Intern
Jointly supervised with Andreas Krause (ETH Zürich).

2025-2026 Benedikt von Querfurth – MSc Thesis and PhD Student
Jointly supervised with Charlotte Bunne (EPFL). Resulting in one publication and two papers under review.

2026 Amine Maazizi – MSc Thesis
Jointly supervised with Antoine Manzanera (ENSTA). Now PhD at ENS. Resulting paper under review.

2026 Ethan Bandasack – MSc Thesis
Jointly supervised with Safouan Taha (Centrale Supélec). Resulting paper under review.

2026 Elisa Billard – MSc Thesis
Jointly supervised with Pietro Liò (Cambridge University), Charlotte Bunne (EPFL). Now PhD at Institute Curie.

2025-2026 Aaron Wenteler – PhD Research Intern
Now Intern at Google Research.

Community Service

2022 - 2025 **Advisory Committee of the Oxford ML Summer School**
AI for Global Goals.

2016 - 2017 **Speaker for Public Relations**
AStA (Student Council) at the University of Mannheim.

2015 - 2016 **Member of the Student Parliament and Deputy Member of the Information Committee**
University of Mannheim. Representative for Refugee Projects.

Fellowships & Awards

Top Reviewer Award – NeurIPS 2026 Dataset & Evaluation track.

Top Paper Award – NeurIPS 2023 Workshop on XAI in Action.

DKFZ International PhD Program Fellowship – Acceptance rate of 2.7% in 2022.

Publications

See also Google Scholar.

** Authors contributed equally.* **(S)** *Students I supervised.*

Key Publications (ordered chronological)

2026 **spora: A Unified Multimodal Dataset for Spatial Proteomics**
Benedikt von Querfurth*(S), Eeshaan Jain*(S), Johann Wenckstern*(S), **Lukas Klein***, Yexiang Cheng, Phil Cheng, Cédric Vincent-Cuaz, Pascal Frossard, Martina Haberecker, Andreas Wicki, Charlotte Bunne. ***NeurIPS** 2026.*

2026 **Explainable AI-based analysis of human pancreas sections identifies traits of type 2 diabetes**
Lukas Klein*, Sebastian Ziegler*, Yanni Morgenroth*, Eyke Schoeniger, Michele Solimena, Felicia Gerst, Robert Wagner, Fabian Isensee, Klaus Maier-Hein, Paul Jäger. ***Nature Communications.***
- Contributed talk at the NeurIPS 2024 IAI Workshop.
- This work has been covered by GEN.

2023 **Discovering Process Dynamic for Scalable Perovskite Solar Cell Manufacturing with Explainable AI**
Lukas Klein*, Sebastian Ziegler*, Felix Laufer*, Charlotte Debus, Markus Götz, Klaus Maier-Hein, Ulrich W. Paetzold, Fabian Isensee, Paul Jäger. ***Advanced Materials.***
- Contributed talk at the NeurIPS 2023 Workshop on XAI in Action (Top Paper Award).
- This work has been covered by The Independent.

Further Conference and Journal Publications (ordered chronological)

2025 **nnActive: A Framework for Evaluation of Active Learning in 3D Biomedical Segmentation**
Carsten T. Lüth*, Jeremias Traub*, Kim-Celine Kahl, Till J. Bungert, **Lukas Klein**, Lars Krämer, Paul Jäger, Fabian Isensee, Klaus Maier-Hein. *Transactions on Machine Learning Research (**TMLR**).*

2025 **Finally Beating the Random Baseline: A Simple and Effective Active Learning Method for 3D Biomedical Segmentation**
Carsten T. Lüth*, Jeremias Traub*, Kim-Celine Kahl, Till J. Bungert, **Lukas Klein**, Lars Krämer, Paul Jäger, Klaus Maier-Hein, Fabian Isensee. *Transactions on Machine Learning Research (**TMLR**).*

2024 **Navigating the Maze of Explainable AI: A Systematic Approach to Evaluating Methods and Metrics**
Lukas Klein, Carsten Lüth, Udo Schlegel, Till Bungert, Mennatallah El-Assady, Paul Jäger. *Neural Information Processing Systems (**NeurIPS**) 2024.*

2024 **Enhancing predictive imaging biomarker discovery through treatment effect analysis**
Shuhan Xiao, **Lukas Klein**, Paul Jäger, Jens Petersen, Klaus Maier-Hein. *IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**) 2025.*

2023 **Navigating the Pitfalls of Active Learning Evaluation: A Systematic Framework for Assessment**
Carsten T. Lüth, Till J. Bungert, **Lukas Klein**, Paul Jäger. ***NeurIPS** 2023.*

2023 **Semi-supervised automated Gleason Grading on WSI**
Maximilian Fischer*, Lars Krämer*, Sebastian Ziegler*, **Lukas Klein***, Paul Jäger, Marco Nolden, Peter Neher, Fabian Isensee, Klaus Maier-Hein. *European Congress on Digital Pathology (ECDP) 2024.*

2023 **A Call to Reflect on Evaluation Practices for Failure Detection in Image Classification**
Paul Jäger, Carsten T. Lüth, **Lukas Klein**, Till J. Bungert. ***ICLR** 2023 **(Oral, Top 1% Paper).***

2022 **Improving Explainability of Disentangled Representations using Multipath-Attribution Mappings**
Lukas Klein, João Carvalho, Mennatallah El-Assady, Paolo Penna, Joachim M. Buhmann, Paul Jäger. *Medical Imaging with Deep Learning (**MIDL**) 2022.*

Preprints and Under Review (ordered chronological)

2026 **Single-Cell Co-Expression NEtworks reveal Actionable Resistance Mechanisms to Targeted and Immunotherapy Treatments in BRAF-mutated Melanoma**
Eva Camarillo-Retamosa, Lars Bosshard, Parisa Rahimzadeh, Benedict Gosztonyi, Lukas Klein, […], Michael Prummer, Andreas Wicki. ***Cell Reports** (under review).*

2026 **Clinical Robustness: Rethinking Foundation Models for Diagnostic Gleason Grading**
Xinmi Huo, Eric Monzon, Kokhaur Ong, […], **Lukas Klein**, […], Dalong Cao, Li Liu, Huakang Tu, Soo Yong Tan, Huanfen Zhao. *Medical Image Analysis (**MIA**) (under review).*

2026 **Foundation Model-based Simulation of Spatiotemporal Tissue Responses to Therapy**
Amine Maazizi*(S), **Lukas Klein***, Benedikt von Querfurth*(S), Johann Wenckstern (S), Justine Epiney, Anne Cairoli, Albert Santamaria-Martínez, Elisa Oricchio, Charlotte Bunne. *Under submission.*

2024 **Interactive Semantic Interventions for VLMs: Breaking VLMs with Human Ingenuity**
Lukas Klein*, Kenza Amara*, Carsten Lüth, Antonio Rodríguez, Hendrik Strobelt, Mennatallah El-Assady, Paul Jäger. *NeurIPS 2024 Safe Generative AI Workshop* and *Causality & Large Models Workshop.*

2022 **From Correlation to Causation: Formalizing Interpretable Machine Learning as a Statistical Process**
Lukas Klein, Mennatallah El-Assady, Paul Jäger. *IJCAI 2022 Workshop on XAI.*

Languages

Languages: German (Native), English (Fluent)

Programming Languages: Python, R, Bash, Git, LaTeX, SQL